

→ Built-in Security Features in Android.

Android includes multiple layers of security designed to protect users, their data, and applications. Some of the key built-in security measures are:

* Application Sandbox :-

Every android app runs in its own separate, isolated environment, which restricts it from accessing the data or system resources of other apps without proper authorization. This containment reduces the potential impact of any harmful app.

* Encryption :-

Android employs full-disk encryption or file-based encryption to secure user data when the device is inactive, ensuring the information remains unreadable without valid authentication, such as password or fingerprint.

⇒ Permissions System -

Android follows a permission-based model, requiring apps to request access to sensitive data or system features such as camera, location, or contacts. Users must manually approve these permissions before the app can use them.

⇒ Security Updates

Google regularly releases Android security patches to address vulnerabilities. Devices that are actively supported receive these updates on a monthly or quarterly basis.

⇒ Biometric Authentication

Android supports secure biometric authentication methods including fingerprint, facial recognition, and iris scanning, to unlock the device or verify sensitive actions such as mobile payments.

⇒ App Signing

Every Android application must be digitally signed by its developer, ensuring that any updates originate from the original source and have not been altered or compromised.